

Von Kármán Cylinder Flow (SimFlow 5.0, PISO) – Reynolds-Based Setup

Physics & numerics:

- Physics: Incompressible · Transient · Laminar
- Solver: PISO (pisoFoam)
- Schemes: Backward (time), LUST (convection)
- 2D: front/back = Empty
- Run controls:
Simulation time = 5 s, $\Delta t = 1e-3$ s, Output interval = 0.025 s (= 25 steps)

Parameters & etc.

1. Parameters

- $d = 0.01$ (m)
- $\nu = 1.51e-05$ (m²/s)
- Re (20 or 100)
- $u = Re \cdot \nu / d$

2. Transport Properties

- Kinematic viscosity = ν

3. Boundary Conditions → Inlet

- Velocity Inlet
- Reference Value = u
- Outlet = 0 Pa, Cylinder = no-slip, Slip/Symmetry, front/back = Empty

4. Monitors / Function Objects

- forceCoeffs $U_\infty = u$
- vorticity (Full Time Range → Calculate)

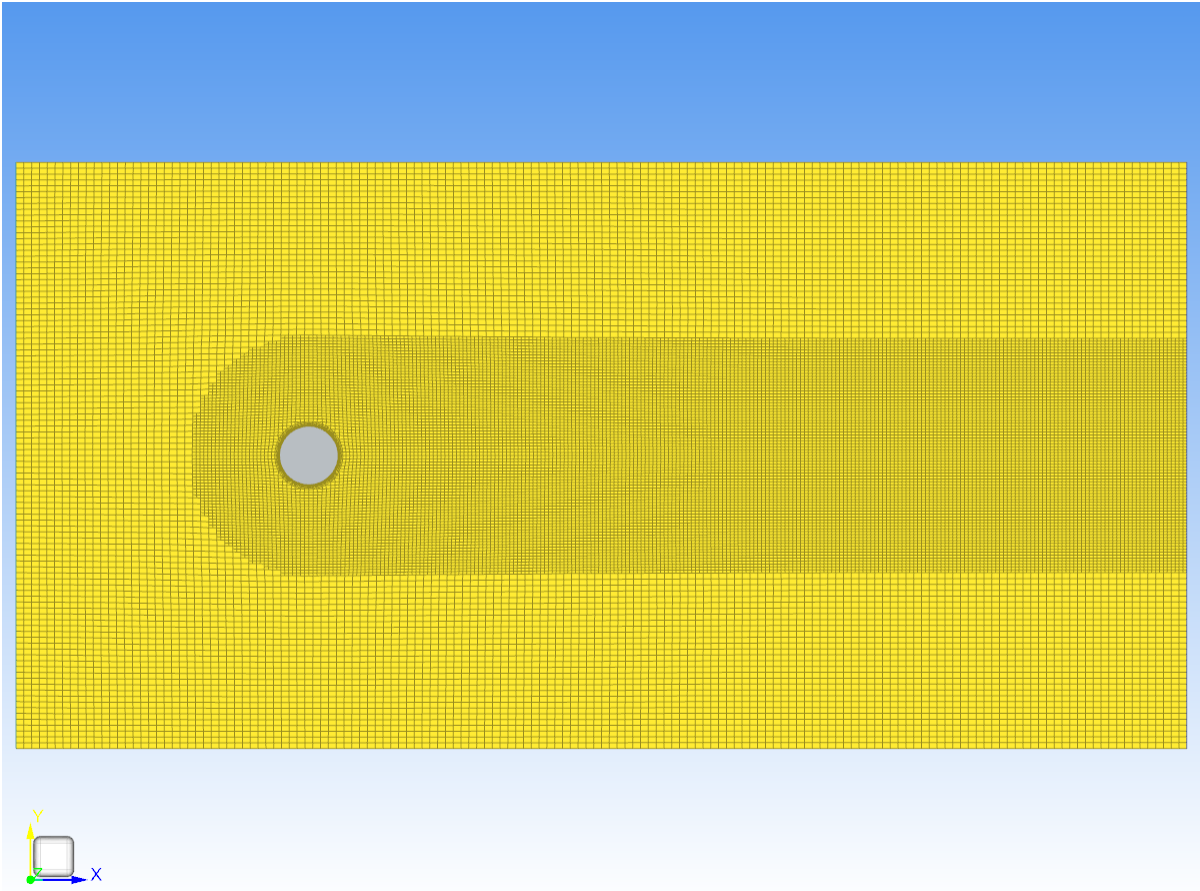
Numbers (with $d = 0.01$, $\nu = \nu$)

$$U = \frac{Re \cdot \nu}{d}$$

- Re=20 → $U = 0.0302$ m/s
- Re=100 → $U = 0.1510$ m/s

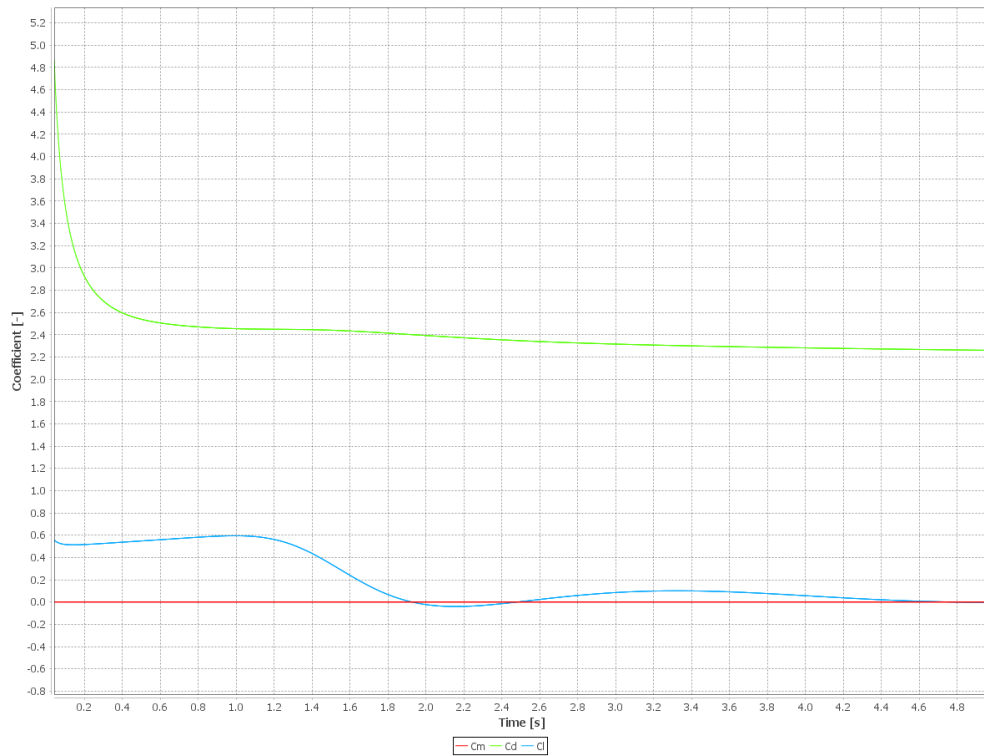
Geometry & Mesh

- Min [m]: (X, Y) = (-0.05, -0.05)
- Max [m]: (X, Y) = (0.15, 0.05)
- Mesh Division = 150 / 100

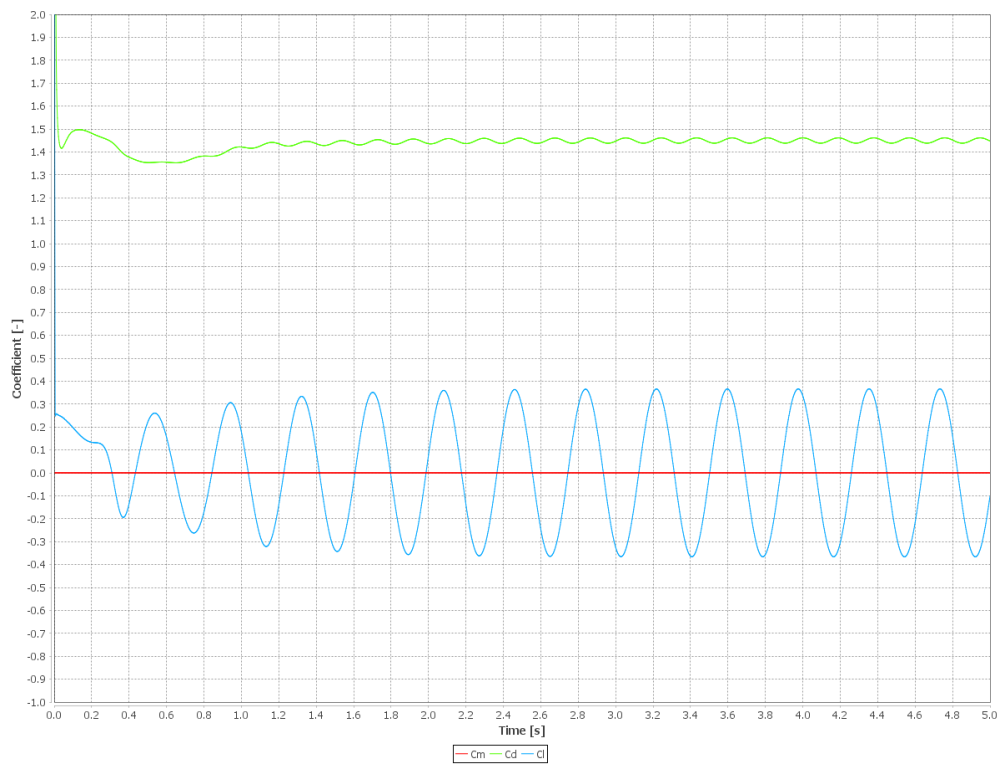


Results – Oscillation Plot

- $Re = 20$



- $Re = 100$

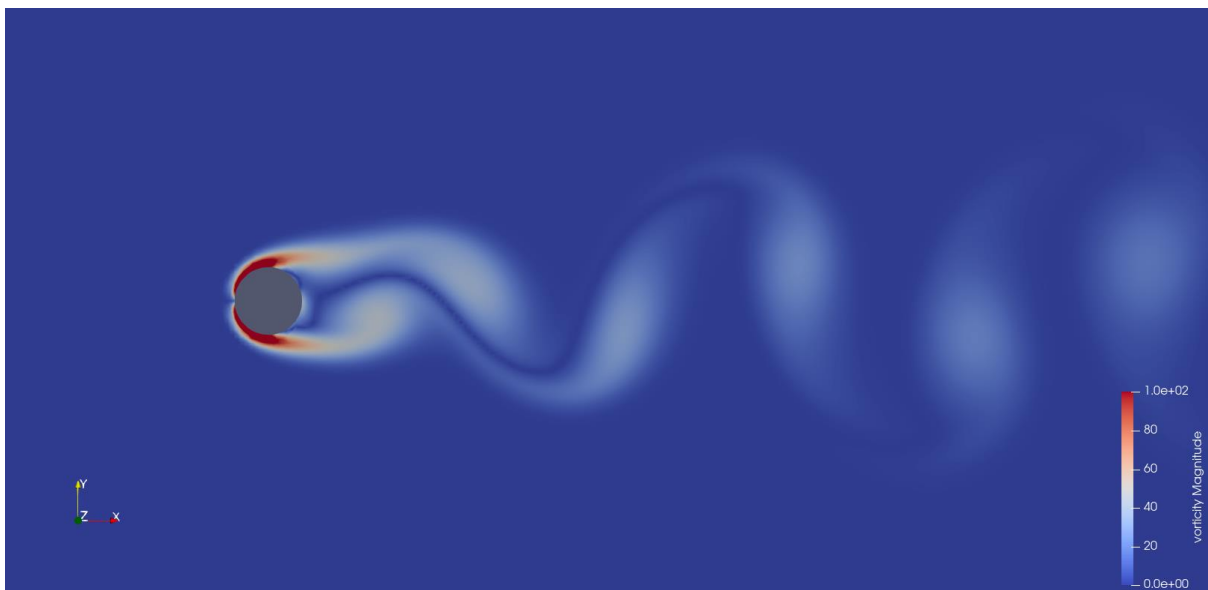


Results – Vorticity Contour Plot

- $Re = 20, t = 5 \text{ s}$



- $Re = 100, t = 5 \text{ s}$



Kármán vortex street/

- └─ Reynolds_20/
 - | └─ VTK/
 - | | └─ Reynolds20.vtm.series # time-series index
 - | | └─ Reynolds20_0.vtm
 - | | └─ Reynolds20_25.vtm
 - | | └─ ...
 - | | └─ Reynolds20_4975/
 - | | └─ Reynolds20_5000/
 - | └─ probe_centerline_u.csv # t, Ux, Uy, |U|, p, vorticityZ, X, Y
 - | └─ metadata.txt
 - | └─ Reynolds20_oscillation_data.csv
 - | └─ Reynolds20_oscillation_plot.png
 - | └─ Reynolds20_vorticity.mp4
 - | └─ Reynolds20_vorticity_2s.png
 - | └─ Reynolds20_vorticity_5s.png
 - | └─ solver.txt
- └─ Reynolds_100/
 - | └─ VTK/
 - | | └─ Reynolds100.vtm.series
 - | | └─ Reynolds100_0.vtm
 - | | └─ Reynolds100_25.vtm
 - | | └─ ...
 - | | └─ Reynolds100_**** /
 - | | └─ ...
 - | └─ probe_centerline_u.csv
 - | └─ metadata.txt
 - | └─ Reynolds100_oscillation_data.csv
 - | └─ Reynolds100_oscillation_plot.png
 - | └─ Reynolds100_vorticity.mp4
 - | └─ Reynolds100_vorticity_2s.png
 - | └─ Reynolds100_vorticity_5s.png
- └─ README.md